ANALYSTLAB AFRICA

Executive Summary: Hospital Operations, Cost & Patient Flow Intelligence

Week 3 | Track A: Data Analytics | Prepared by Kukoyi Zainab Omotola | November 2026

Purpose and scope

This summary reports what a Power BI dashboard built on the supplied hospital datasets shows about encounter volumes, encounter classes, claim costs, payer coverage and procedure use. It is written for hospital operations and finance leaders. The data covers one hospital from 2 January 2011 to 5 February 2022: 27,891 encounters, 974 patients, 47,701 procedure records, 10 payers and total claim cost of about $101.5M (an average of $3,640 per encounter).

Key findings

- **Volume.** Encounters rose to a peak of 3,885 in 2014, settled at about 2,200 to 2,500 a year from 2015 to 2020, and rose again to 3,530 in 2021. February 2014 alone has 1,179 encounters, against about 200 in a typical month. 2022 holds only 220 encounters because the data ends on 5 February.
- **Encounter classes.** Ambulatory (45.0%) and outpatient (22.6%) make up 67.5% of encounters but 49.6% of claim cost. Urgent care is 13.1% of encounters but 23.0% of claim cost, with an average claim of $6,369. Inpatient has the highest average claim ($7,761) and accounts for 58 of the 75 encounters (0.27%) that lasted more than 24 hours. A few of those 75 are extreme (13 last over 30 days, and the longest about 5 years), so the measure mixes long stays with long-running records.
- **Payer coverage.** Payers are recorded as covering $31.1M, or 30.6% of claim cost. 13,586 encounters (48.7%) have no recorded payer coverage, and they carry about $63.1M (62.2%) of claim cost. 8,807 of them are with the NO_INSURANCE payer. The other 4,779 are with named insurers.
- **Coverage differs sharply by payer.** Humana, Aetna, UnitedHealthcare, Cigna and Anthem record 0.0% to 0.2% of their claim cost as payer coverage. Medicare, Medicaid and Dual Eligible record 78% to 94%. Average claim cost by payer ranges from $1,696 (Dual Eligible) to $6,205 (Medicaid).
- **Procedures.** The 47,701 procedure records cover 163 descriptions. The ten most frequent make up 63% of records, and most are routine assessments and screenings with a flat base cost of $431. Electrical cardioversion has both high cost and high volume (1,383 records, average base cost $25,903, total about $35.8M), the largest cost concentration. 47.4% of encounters have no procedure record, and procedure start times fall outside their encounter's time window in 51% of cases, so procedures are dated by their encounter.

Recommendations

1. Check how payer coverage is recorded for five private insurers

Finding: Humana, Aetna, UnitedHealthcare, Cigna and Anthem show almost no recorded coverage, while Medicare, Medicaid and Dual Eligible show 78% to 94%. 4,779 zero-coverage encounters belong to named insurers.

Proposed action: Review a sample of claims for these five payers to confirm whether coverage is being recorded correctly in the source system.

Intended benefit: A reliable picture of how much each payer funds, and a clean basis for payer-level reporting.

Limitation / information needed: The data does not show whether the coverage is unrecorded, unpaid or genuinely zero. Billing and remittance records are needed to say which.

2. Quantify the exposure from encounters with no recorded payer coverage

Finding: 13,586 encounters (48.7%) have zero recorded coverage and carry about $63.1M (62.2%) of claim cost. 8,807 are with the NO_INSURANCE payer, which alone carries $49.3M.

Proposed action: Add a field showing who pays when there is no payer coverage (for example patient, charity or write-off) and track the cost of these encounters over time.

Intended benefit: Finance can plan for this cost with evidence instead of assumption.

Limitation / information needed: The data has no payment or write-off fields, so this cost cannot be called a loss. Payment data is needed.

3. Prioritise cost review for urgent care and inpatient encounters

Finding: Urgent care is 13.1% of encounters but 23.0% of claim cost, and inpatient has the highest average claim ($7,761) and nearly all long stays.

Proposed action: Review what drives claim cost in these two classes, starting with the procedures attached to them.

Intended benefit: Focuses cost-control effort where cost per encounter is highest.

Limitation / information needed: How claim cost is built up is not documented, and class differences may reflect case mix. No clinical conclusions can be drawn from this data.

4. Validate the cost of electrical cardioversion and other high-cost procedures

Finding: Electrical cardioversion has 1,383 records, an average base cost of $25,903 and about $35.8M in total, the largest concentration. Rare procedures such as ICU admission average over $200,000 but have only 5 records.

Proposed action: Check that the recorded base costs for these procedures are correct and consistent, and review whether they are routinely priced as expected.

Intended benefit: Confirms where the largest procedure costs sit and removes the risk of acting on pricing errors.

Limitation / information needed: Base cost is not claim cost: procedure base costs total about $105.5M, more than total claim cost, so the two must not be added. Averages for procedures with under 20 records are unstable.

5. Improve data completeness and explain unusual volume

Finding: 47.4% of encounters have no procedure record, procedure timestamps fall outside their encounter's time window in 51% of cases, a few encounters last for months or years, reason fields are blank for 70% of encounters, currency is not stated, and February 2014 (1,179 encounters) is unexplained.

Proposed action: Confirm why February 2014 is so high and why some encounters last for months or years, align procedure timestamps with encounter times, record reasons and procedures consistently, and state the currency in the data dictionary.

Intended benefit: Makes volume trends, capacity planning and procedure analysis more dependable.

Limitation / information needed: Until the cause of the 2014 and 2021 peaks is known, they should not be used as a planning baseline.

Data quality and limitations

- The data has no duplicate records and no orphan keys. No rows were removed or values replaced.
- It covers one hospital, 2022 is a partial year, and records appear to be synthetic or anonymised.
- Zero payer coverage may mean uninsured, unrecorded or unpaid; the data does not say which.
- Procedure timestamps do not align with encounter times in 51% of cases, so procedures are dated by their encounter. The over-24-hours measure includes a few extreme durations (13 encounters exceed 30 days).

- The data holds no clinical outcomes, so no clinical claims or treatment recommendations are made.
- Patient names, addresses and other identifiers were removed before loading and do not appear in the dashboard.

Validation

Headline totals (encounters, claim cost, payer coverage, procedure count and the rates derived from them) were calculated directly from the source files and compared with the dashboard. Each check, the independent Excel and SQL results, and the filter tests are recorded in the validation table in KPI_Definitions.xlsx. KPI definitions are documented in the same workbook so another analyst can reproduce them.